

# Equity Cash Return and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Equity Cash Return (ECR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on ECR achieves an annualized gross (net) Sharpe ratio of 0.53 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (29) bps/month with a t-statistic of 2.66 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 19 bps/month with a t-statistic of 2.28.

# 1 Introduction

Financial markets efficiently incorporate information about firms’ production decisions and investment opportunities into asset prices, yet significant cross-sectional variation in expected returns persists even after controlling for standard risk factors. This variation reflects differences in firms’ exposure to aggregate productivity shocks and their ability to adjust production in response to changing economic conditions. We examine Equity Cash Return (ECR), defined as operating cash flow scaled by market equity, as a signal that captures firms’ marginal productivity of capital and their exposure to systematic investment risks. The production-based asset pricing literature establishes that firms facing higher adjustment costs or greater exposure to productivity shocks require higher risk premia to compensate investors [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). ECR directly measures the cash flow productivity of deployed equity capital, providing a market-based assessment of firms’ production efficiency that incorporates both current profitability and future investment frictions.

In a production economy, firms’ optimal investment decisions depend on the marginal product of capital relative to adjustment costs and stochastic discount factors [Cochrane \(1991\)](#); [Zhang \(2005\)](#). ECR captures this marginal productivity by measuring the cash flows generated per unit of market-valued equity capital. Firms with high ECR demonstrate efficient conversion of equity capital into operating cash flows, indicating either high productivity or low adjustment costs in their production functions. The q-theory of investment predicts that firms with higher marginal products of capital face steeper adjustment cost curves, making their values more sensitive to aggregate productivity shocks [Liu et al. \(2009\)](#). These firms cannot easily adjust their capital stock in response to negative shocks, creating operating leverage that amplifies their systematic risk exposure.

The production-based framework further implies that ECR proxies for firms’ exposure to investment-specific technology shocks and time-varying risk premia [Pa-](#)

panikolaou and Kogan (2011); Yang and Kogan (2013). High ECR firms operate with capital-intensive production technologies that generate substantial cash flows but face significant reinvestment requirements to maintain productivity. During periods of heightened aggregate uncertainty or deteriorating investment opportunities, these firms experience larger valuation impacts because their high cash flow productivity cannot be easily redeployed. The irreversibility of capital investments combined with time-varying discount rates creates a systematic risk factor that prices the cross-section of returns Barro and Ursúa (2008).

Adjustment costs in production create a wedge between firms' optimal and actual capital stocks, generating predictable variation in expected returns Cooper and Priestley (2009); Lin and Zhang (2013). ECR measures the efficiency of this capital deployment, with high values indicating firms operating closer to their optimal scale despite facing adjustment frictions. These firms exhibit greater sensitivity to aggregate investment shocks because marginal changes in productivity or discount rates have amplified effects on their valuations. The production-based model predicts that sorting on ECR creates portfolios with differential loadings on aggregate investment and productivity factors, generating the cross-sectional return predictability we document.

We find that a value-weighted long/short trading strategy based on ECR achieves an annualized gross Sharpe ratio of 0.53, with the high ECR portfolio earning 93 basis points per month compared to 41 basis points for the low ECR portfolio. The strategy generates a monthly average abnormal return of 24 basis points relative to the Fama-French five-factor model augmented with momentum (FF6), with a t-statistic of 2.66. This performance remains economically and statistically significant after accounting for transaction costs, with the net Sharpe ratio of 0.50 exceeding 99% of anomalies in the factor zoo. The strategy's alpha relative to the FF6 model plus six closely related anomalies equals 19 basis points per month with a t-statistic

of 2.28, confirming that ECR contains unique information about expected returns.

The ECR strategy exhibits significant exposure to the profitability factor (RMW) with a loading of 0.76 (t-statistic of 18.88), consistent with the production-based interpretation that high ECR firms face greater operating leverage and systematic risk. Importantly, the strategy performs strongly among large-cap stocks, generating an average return of 47 basis points per month with a t-statistic of 3.17 for firms above the 80th NYSE size percentile. The FF6 alpha for this large-cap subsample equals 14 basis points per month, demonstrating that ECR’s predictive power extends beyond small, illiquid stocks where many anomalies concentrate.

Robustness tests confirm that ECR’s return predictability persists across alternative portfolio construction methodologies and withstands controls for related signals. Using different sorting breakpoints and weighting schemes, we obtain alphas ranging from 12 to 76 basis points per month, with the majority exhibiting t-statistics exceeding 2.0. Fama-MacBeth regressions controlling for six closely related anomalies yield a coefficient on ECR of 0.15 with a t-statistic of 2.16, while spanning tests produce an alpha of 19 basis points per month (t-statistic of 2.28) after controlling for these same factors. The ECR signal’s gross alpha ranks between the 72nd and 95th percentiles across standard factor models, placing it among the strongest documented return predictors.

Our study extends the production-based asset pricing literature by demonstrating that market-based measures of capital productivity contain significant information about cross-sectional return variation. While [Fama and French \(2006\)](#) and [Novy-Marx \(2013\)](#) document the importance of profitability for expected returns, we show that scaling operating cash flows by market equity rather than book values better captures firms’ exposure to systematic investment risks. ECR differs from the operating profitability measure in [Fama and French \(2015\)](#) by using market values that incorporate forward-looking information about investment opportunities and adjust-

ment costs. Unlike the return on assets measure studied in [Chen et al. \(2011\)](#), ECR focuses specifically on cash generation efficiency, which more directly reflects the marginal product of capital in production-based models.

We contribute methodologically by applying the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#) to evaluate ECR’s incremental contribution to the investment opportunity set. Our analysis demonstrates that ECR expands the mean-variance efficient frontier even after accounting for transaction costs and controlling for 212 documented anomalies. The signal’s ability to predict returns among large-cap stocks distinguishes it from many profitability-based anomalies that concentrate in smaller, less liquid segments of the market. This finding has important implications for institutional investors who face capacity constraints when implementing factor strategies.

Our results provide new evidence supporting production-based theories of asset pricing by identifying a theoretically motivated signal that captures firms’ heterogeneous exposure to aggregate productivity and investment shocks. The strong performance of ECR across different market capitalizations and time periods suggests that production frictions and adjustment costs create persistent sources of systematic risk in equity returns. These findings contribute to the growing literature linking firm production decisions to asset prices, as exemplified by [Lin and Zhang \(2013\)](#) and [Hou et al. \(2015\)](#), while offering practitioners a robust signal for constructing factor portfolios. The economic magnitude and statistical significance of ECR’s return predictability, combined with its theoretical foundations in production economics, establish it as an important addition to the cross-sectional asset pricing literature.

## 2 Data

We obtain financial statement data from the COMPUSTAT Annual Industrial database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data necessary to construct the Equity Cash Return signal, spanning several decades of financial reporting. We focus on U.S.-listed companies with common shares traded on major exchanges (NYSE, AMEX, and NASDAQ). Equity Cash Return signal requires two key accounting variables from COMPUSTAT. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a company's core business operations during the fiscal year, as reported in the statement of cash flows. This measure captures the actual cash inflows and outflows from operating activities, including cash received from customers, cash paid to suppliers and employees, and other operating cash transactions. Stockholders' equity (SEQ) represents the book value of shareholders' residual claim on the company's assets after deducting all liabilities, as reported on the balance sheet. This measure reflects the cumulative net assets attributable to equity holders, including contributed capital and retained earnings. We construct the Equity Cash Return signal by dividing operating activities net cash flow (OANCF) by stockholders' equity (SEQ) for each firm-year observation. This ratio measures the cash-generating efficiency of shareholders' invested capital, capturing how effectively a firm converts its equity base into operating cash flows. We calculate the signal annually using fiscal year-end values from firms' financial statements. To ensure data quality, we require non-missing values for both OANCF and SEQ, and we exclude observations where stockholders' equity is zero or negative to avoid undefined or economically meaningless ratios. The resulting measure provides a cash-based return metric that complements traditional accrual-based profitability measures such as return on equity.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECR signal. Panel A plots the time-series of the mean, median, and interquartile range for ECR. On average, the cross-sectional mean (median) ECR is 0.04 (0.13) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ECR data. The signal’s interquartile range spans -0.24 to 0.29. Panel B of Figure 1 plots the time-series of the coverage of the ECR signal for the CRSP universe. On average, the ECR signal is available for 7.39% of CRSP names, which on average make up 7.94% of total market capitalization.

### 4 Does ECR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ECR portfolio and sells the low ECR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ECR strategy earns an average return of 0.52% per month with a t-statistic of 3.20. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.24% to 0.76% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.76,

with a t-statistic of 18.88 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,746 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 41 bps/month with a t-statistics of 2.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for seventeen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 22-50bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.27. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the ECR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECR, as well as average returns and alphas for long/short trading ECR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the ECR strategy achieves an average return of 47 bps/month with a t-statistic of 3.17. Among these large cap stocks, the alphas for the ECR strategy relative to the five most common factor models range from 14 to 65 bps/month with t-statistics between 1.21 and 4.73.

## 5 How does ECR perform relative to the zoo?

Figure 2 puts the performance of ECR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the ECR strategy falls in the distribution. The ECR strategy’s gross (net) Sharpe ratio of 0.53 (0.50) is greater than 92% (99%) of anomaly Sharpe ratios,

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECR strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the ECR strategy would have yielded \$6.70 which ranks the ECR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECR strategy would have yielded \$5.88 which ranks the ECR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECR relative to those. Panel A shows that the ECR strategy gross alphas fall between the 72 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECR strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECR ranks between the 93 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

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<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

## 6 Does ECR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ECR with 210 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECR or at least to weaken the power ECR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ECR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ECR}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ECR}ECR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ECR,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECR. Stocks are finally grouped into five ECR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ECR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECR signal in these Fama-MacBeth regressions exceed 0.07, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on ECR is 2.16.

Similarly, Table 5 reports results from spanning tests that regress returns to the ECR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECR strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.24, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECR trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.28.

## 7 Does ECR add relative to the whole zoo?

Finally, we can ask how much adding ECR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ECR signal.<sup>4</sup> We consider one different methods for combining signals.

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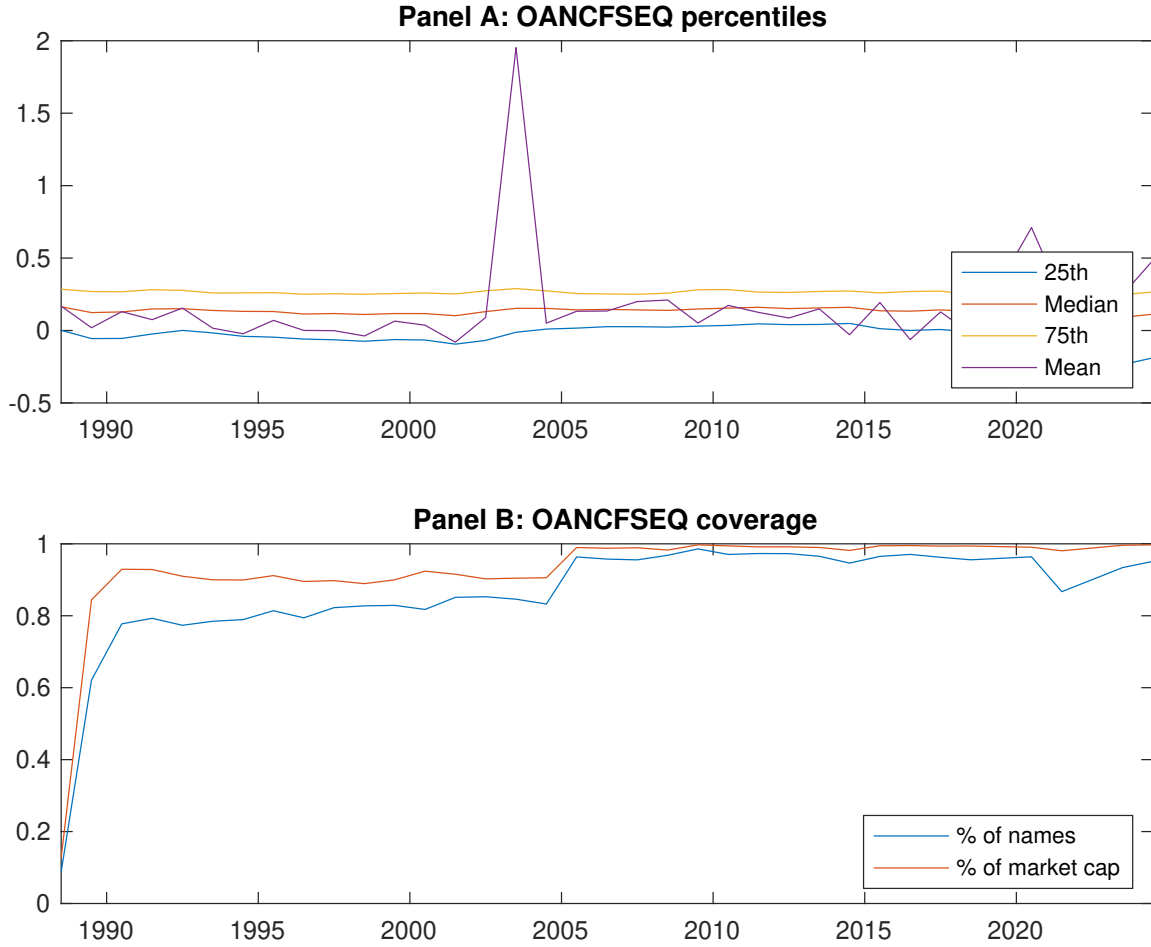
<sup>4</sup>We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECR is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes ECR grows to \$44.64.

## 8 Conclusion

This study provides compelling evidence for the predictive power of Equity Cash Return (ECR) as a robust signal for cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ECR generates economically and statistically significant abnormal returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy based on ECR delivers an impressive annualized Sharpe ratio of 0.53 gross (0.50 net), indicating strong risk-adjusted performance that remains viable after accounting for transaction costs. Most notably, the signal maintains statistical significance with a monthly alpha of 19 basis points ( $t$ -statistic = 2.28) even when confronted with a comprehensive set of twelve factors, including the Fama-French five factors, momentum, and six conceptually related strategies from the factor zoo. This resilience suggests that ECR captures unique information about firm fundamentals not fully reflected in existing pricing models or related profitability and financing measures. The practical implications of these findings are substantial for both academic researchers and practitioners. For asset managers, ECR represents a potentially valuable addition to multi-factor strategies, offering diversification benefits given its ability to generate

alpha beyond established factors. The signal’s economic significance after transaction costs suggests real-world implementability, though practitioners should carefully consider market impact and capacity constraints in live trading environments. However, several limitations warrant consideration. First, while our analysis follows best practices for anomaly testing, the true out-of-sample performance of ECR remains to be seen as markets evolve and adapt to newly documented anomalies. Second, the study focuses primarily on U.S. equity markets, and the international robustness of ECR requires further investigation. Future research should explore several promising avenues. Cross-market validation in international developed and emerging markets would strengthen the generalizability of our findings. Additionally, examining the economic mechanisms underlying ECR’s predictive power through detailed fundamental analysis and exploring potential interactions with macroeconomic conditions could provide deeper insights into when and why the signal performs. Researchers might also investigate whether ECR’s effectiveness varies across different market regimes or firm characteristics, potentially uncovering conditional relationships that could enhance implementation strategies. Finally, combining ECR with machine learning techniques or examining its role in explaining other asset pricing puzzles represents fertile ground for advancing our understanding of cross-sectional return predictability.



**Figure 1:** Times series of ECR percentiles and coverage. This figure plots descriptive statistics for ECR. Panel A shows cross-sectional percentiles of ECR over the sample. Panel B plots the monthly coverage of ECR relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ECR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Excess returns and alphas on ECR-sorted portfolios                       |                   |                  |                  |                  |                  |                   |
|-----------------------------------------------------------------------------------|-------------------|------------------|------------------|------------------|------------------|-------------------|
|                                                                                   | (L)               | (2)              | (3)              | (4)              | (H)              | (H-L)             |
| $r^e$                                                                             | 0.41<br>[1.42]    | 0.59<br>[2.60]   | 0.68<br>[3.19]   | 0.78<br>[3.91]   | 0.93<br>[4.53]   | 0.52<br>[3.20]    |
| $\alpha_{CAPM}$                                                                   | -0.53<br>[-4.85]  | -0.18<br>[-2.70] | -0.04<br>[-0.62] | 0.10<br>[1.82]   | 0.23<br>[4.06]   | 0.76<br>[5.09]    |
| $\alpha_{FF3}$                                                                    | -0.49<br>[-5.54]  | -0.19<br>[-3.29] | -0.06<br>[-0.99] | 0.10<br>[1.77]   | 0.23<br>[4.67]   | 0.71<br>[6.09]    |
| $\alpha_{FF4}$                                                                    | -0.42<br>[-4.84]  | -0.18<br>[-3.04] | -0.04<br>[-0.71] | 0.13<br>[2.40]   | 0.20<br>[4.03]   | 0.62<br>[5.33]    |
| $\alpha_{FF5}$                                                                    | -0.19<br>[-2.66]  | -0.08<br>[-1.43] | -0.06<br>[-1.03] | 0.03<br>[0.49]   | 0.11<br>[2.43]   | 0.30<br>[3.27]    |
| $\alpha_{FF6}$                                                                    | -0.15<br>[-2.13]  | -0.08<br>[-1.34] | -0.05<br>[-0.76] | 0.06<br>[1.08]   | 0.09<br>[1.98]   | 0.24<br>[2.66]    |
| Panel B: Fama and French (2018) 6-factor model loadings for ECR-sorted portfolios |                   |                  |                  |                  |                  |                   |
| $\beta_{MKT}$                                                                     | 1.09<br>[62.16]   | 0.99<br>[69.02]  | 0.98<br>[62.93]  | 0.94<br>[69.61]  | 1.01<br>[91.09]  | -0.07<br>[-3.35]  |
| $\beta_{SMB}$                                                                     | 0.26<br>[10.33]   | 0.09<br>[4.44]   | 0.09<br>[3.95]   | -0.05<br>[-2.69] | -0.12<br>[-7.62] | -0.38<br>[-11.91] |
| $\beta_{HML}$                                                                     | 0.11<br>[3.45]    | 0.21<br>[8.49]   | 0.09<br>[3.17]   | -0.07<br>[-2.99] | -0.11<br>[-5.80] | -0.22<br>[-5.61]  |
| $\beta_{RMW}$                                                                     | -0.54<br>[-17.01] | -0.20<br>[-7.85] | -0.04<br>[-1.35] | 0.16<br>[6.33]   | 0.22<br>[11.06]  | 0.76<br>[18.88]   |
| $\beta_{CMA}$                                                                     | -0.23<br>[-5.20]  | -0.09<br>[-2.57] | 0.11<br>[2.73]   | 0.04<br>[1.12]   | 0.07<br>[2.51]   | 0.30<br>[5.33]    |
| $\beta_{UMD}$                                                                     | -0.06<br>[-3.81]  | -0.01<br>[-0.52] | -0.03<br>[-1.83] | -0.05<br>[-4.14] | 0.03<br>[3.17]   | 0.09<br>[4.58]    |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                   |                  |                  |                  |                  |                   |
| $n$                                                                               | 1590              | 792              | 590              | 527              | 571              |                   |
| $me$ (\$10 <sup>6</sup> )                                                         | 1746              | 2174             | 2618             | 4507             | 5329             |                   |

**Table 2:** Robustness to sorting methodology & trading costs

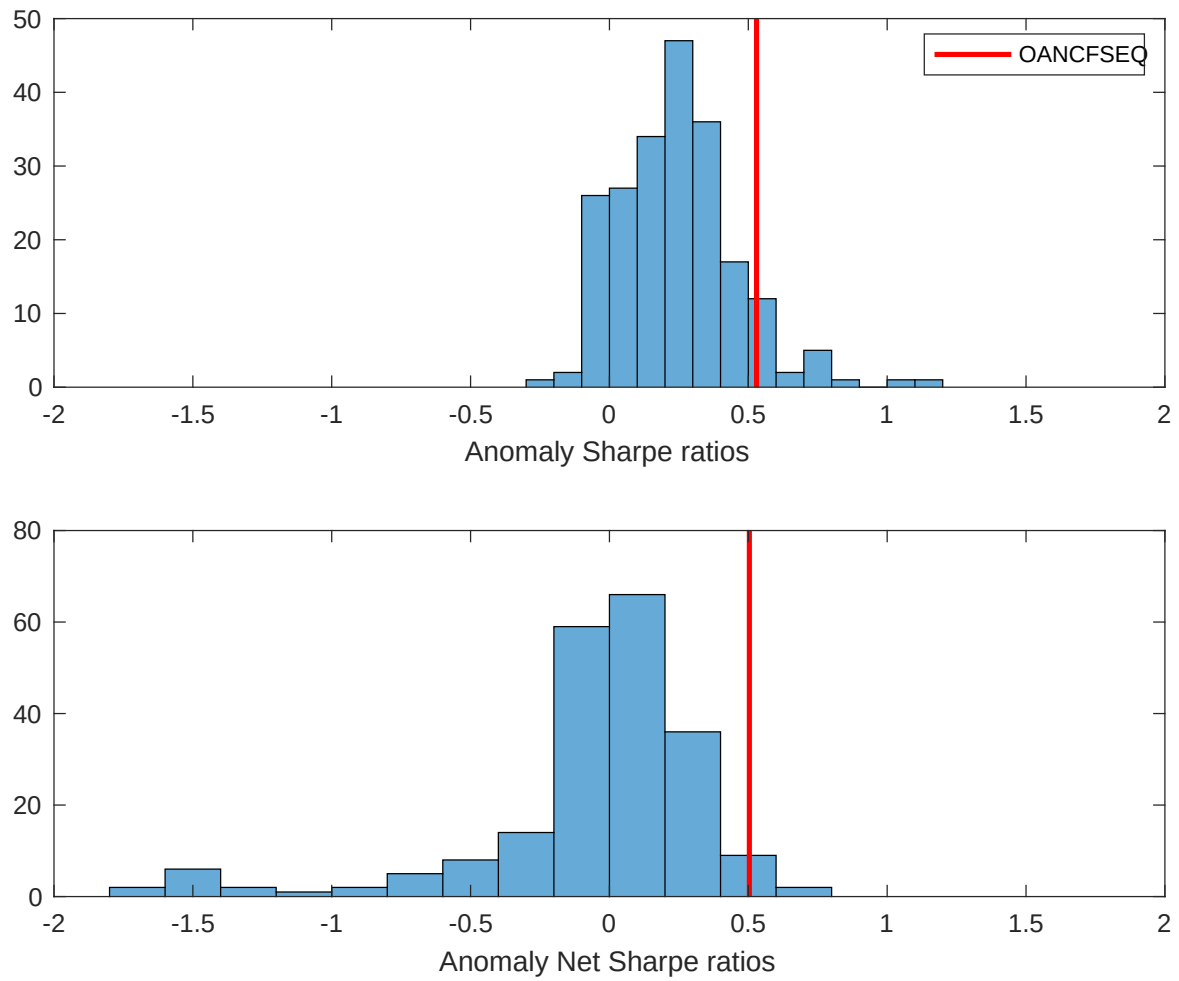
This table evaluates the robustness of the choices made in the ECR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.52<br>[3.20]     | 0.76<br>[5.09]           | 0.71<br>[6.09]          | 0.62<br>[5.33]          | 0.30<br>[3.27]          | 0.24<br>[2.66]          |
| Quintile                                                                 | NYSE   | EW      | 0.41<br>[2.35]     | 0.50<br>[2.91]           | 0.37<br>[2.74]          | 0.26<br>[1.96]          | -0.00<br>[-0.03]        | -0.08<br>[-0.73]        |
| Quintile                                                                 | Name   | VW      | 0.41<br>[2.16]     | 0.66<br>[3.77]           | 0.59<br>[4.18]          | 0.57<br>[3.93]          | 0.26<br>[2.04]          | 0.25<br>[1.93]          |
| Quintile                                                                 | Cap    | VW      | 0.43<br>[2.93]     | 0.60<br>[4.30]           | 0.58<br>[4.95]          | 0.50<br>[4.24]          | 0.17<br>[1.83]          | 0.12<br>[1.29]          |
| Decile                                                                   | NYSE   | VW      | 0.53<br>[2.77]     | 0.79<br>[4.40]           | 0.71<br>[4.70]          | 0.67<br>[4.41]          | 0.24<br>[1.90]          | 0.24<br>[1.84]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.49<br>[3.04]     | 0.74<br>[4.92]           | 0.68<br>[5.79]          | 0.62<br>[5.37]          | 0.34<br>[3.67]          | 0.29<br>[3.27]          |
| Quintile                                                                 | NYSE   | EW      | 0.22<br>[1.27]     | 0.32<br>[1.81]           | 0.21<br>[1.45]          | 0.14<br>[1.03]          |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.37<br>[1.99]     | 0.65<br>[3.63]           | 0.57<br>[3.96]          | 0.55<br>[3.81]          | 0.28<br>[2.16]          | 0.27<br>[2.08]          |
| Quintile                                                                 | Cap    | VW      | 0.41<br>[2.80]     | 0.58<br>[4.07]           | 0.54<br>[4.58]          | 0.48<br>[4.17]          | 0.20<br>[2.19]          | 0.17<br>[1.82]          |
| Decile                                                                   | NYSE   | VW      | 0.50<br>[2.60]     | 0.75<br>[4.14]           | 0.66<br>[4.37]          | 0.64<br>[4.21]          | 0.26<br>[2.04]          | 0.25<br>[1.99]          |

**Table 3:** Conditional sort on size and ECR

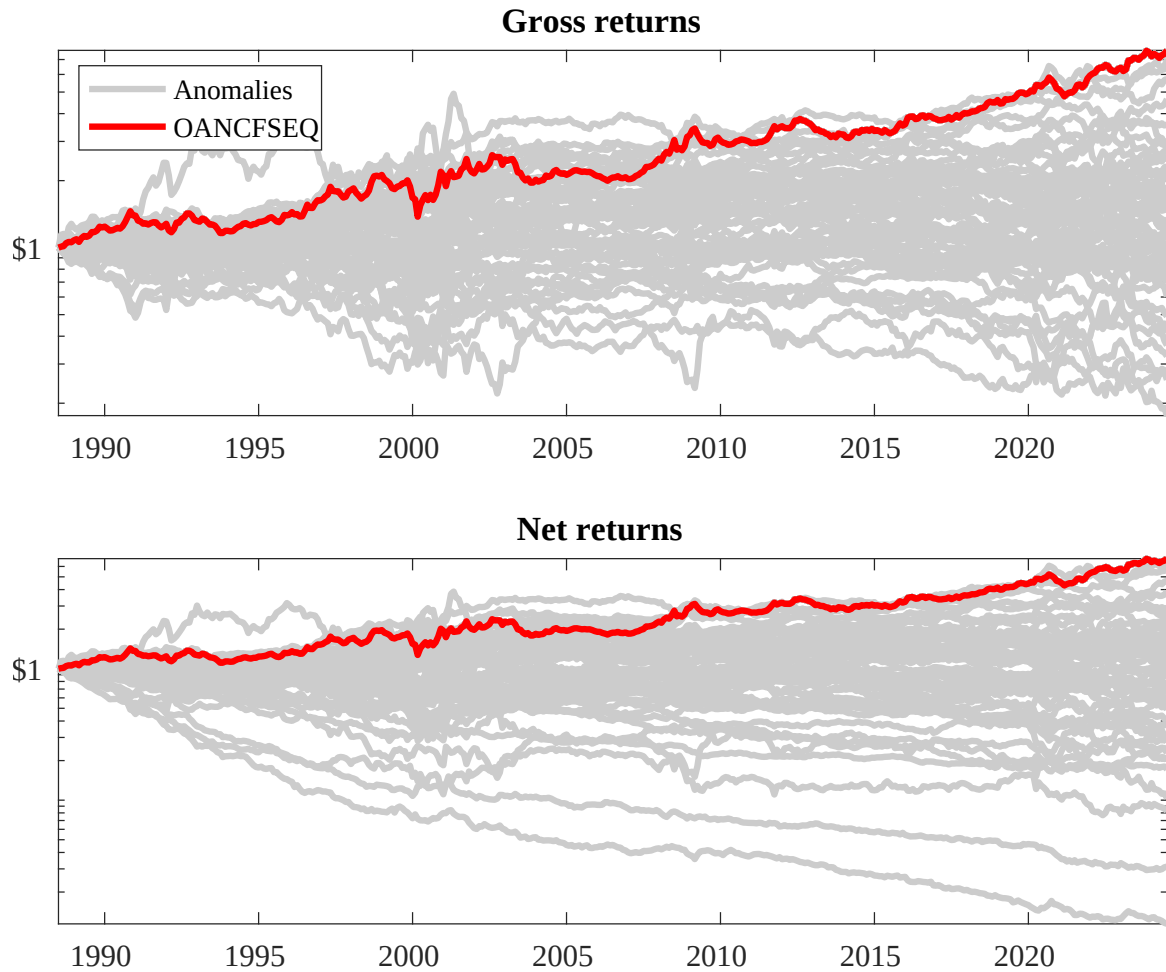
This table presents results for conditional double sorts on size and ECR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECR and short stocks with low ECR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | ECR Quintiles |                |                |                |                | ECR Strategies |                                                    |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.21<br>[0.46] | 0.45<br>[1.21] | 0.64<br>[2.19] | 0.87<br>[3.28] | 1.04<br>[3.43] | 0.84<br>[2.96]                                     | 1.11<br>[4.01]  | 0.89<br>[4.06] | 0.85<br>[3.81] | 0.40<br>[2.08] | 0.38<br>[1.94] |
|                                                                       | (2)           | 0.33<br>[0.83] | 0.55<br>[1.81] | 0.77<br>[2.91] | 0.93<br>[3.57] | 1.06<br>[3.74] | 0.73<br>[3.28]                                     | 0.98<br>[4.58]  | 0.81<br>[4.83] | 0.78<br>[4.61] | 0.34<br>[2.58] | 0.34<br>[2.50] |
|                                                                       | (3)           | 0.33<br>[0.97] | 0.66<br>[2.36] | 0.75<br>[3.02] | 0.96<br>[3.99] | 1.03<br>[3.87] | 0.70<br>[3.67]                                     | 0.89<br>[4.82]  | 0.74<br>[5.08] | 0.69<br>[4.66] | 0.30<br>[2.48] | 0.27<br>[2.23] |
|                                                                       | (4)           | 0.62<br>[2.01] | 0.65<br>[2.63] | 0.82<br>[3.64] | 0.87<br>[3.78] | 0.98<br>[4.03] | 0.36<br>[2.36]                                     | 0.54<br>[3.68]  | 0.44<br>[3.51] | 0.39<br>[3.06] | 0.08<br>[0.76] | 0.05<br>[0.48] |
|                                                                       | (5)           | 0.43<br>[1.70] | 0.60<br>[2.66] | 0.68<br>[3.31] | 0.90<br>[4.40] | 0.90<br>[4.37] | 0.47<br>[3.17]                                     | 0.63<br>[4.42]  | 0.65<br>[4.73] | 0.54<br>[3.94] | 0.21<br>[1.82] | 0.14<br>[1.21] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
| Size quintiles                                                        | ECR Quintiles |                |                |                |                | ECR Quintiles  |                                                    |                 |                |                |                |                |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 438            | 452            | 456            | 458            | 450            | 39                                                 | 46              | 54             | 62             | 61             |                |
|                                                                       | (2)           | 132            | 133            | 133            | 133            | 133            | 88                                                 | 91              | 94             | 97             | 96             |                |
|                                                                       | (3)           | 90             | 90             | 90             | 90             | 90             | 152                                                | 158             | 161            | 164            | 165            |                |
|                                                                       | (4)           | 74             | 74             | 75             | 75             | 74             | 325                                                | 331             | 346            | 347            | 351            |                |
| (5)                                                                   | 66            | 66             | 66             | 66             | 66             | 2075           | 1838                                               | 2300            | 3287           | 3645           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

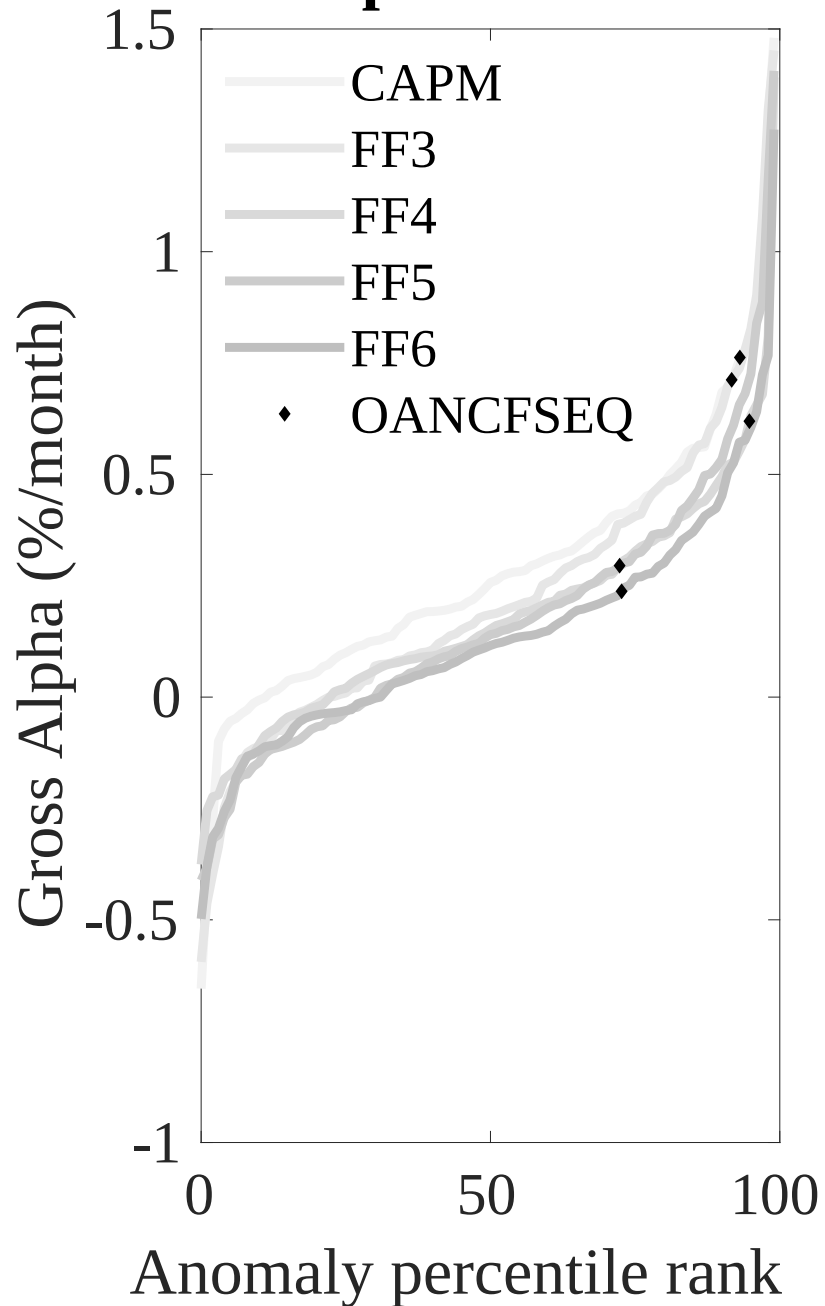
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

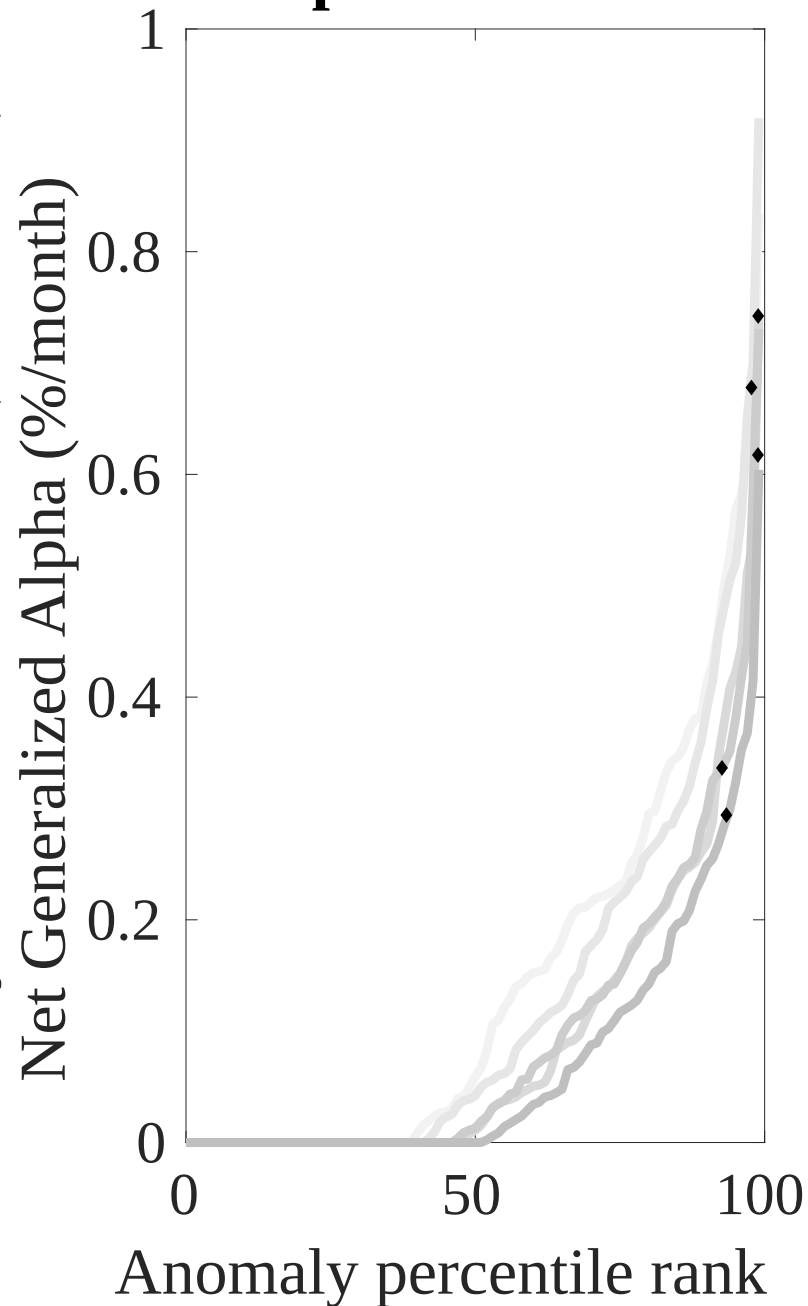
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

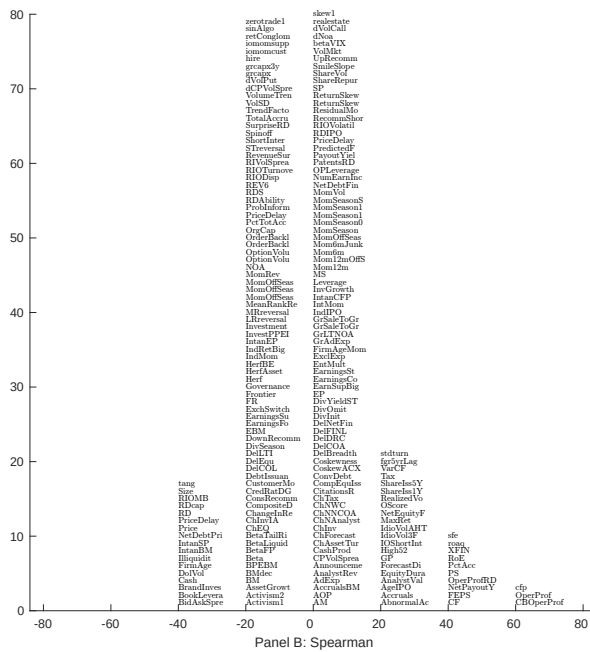
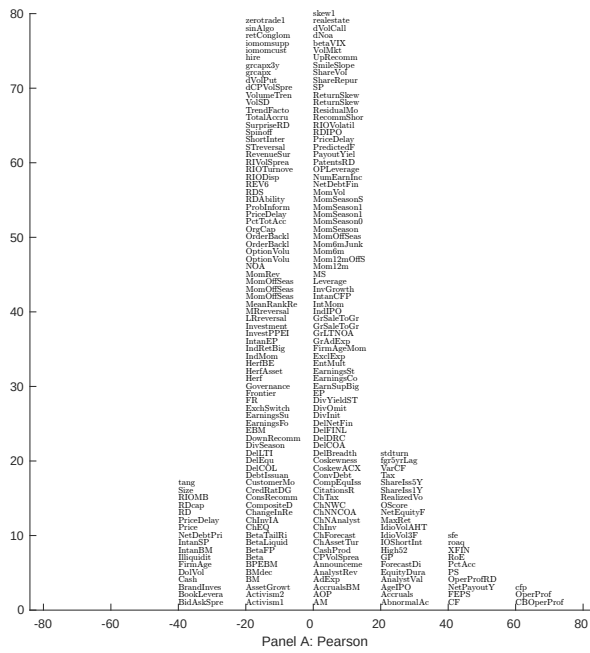
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

## Net Alpha Percentiles

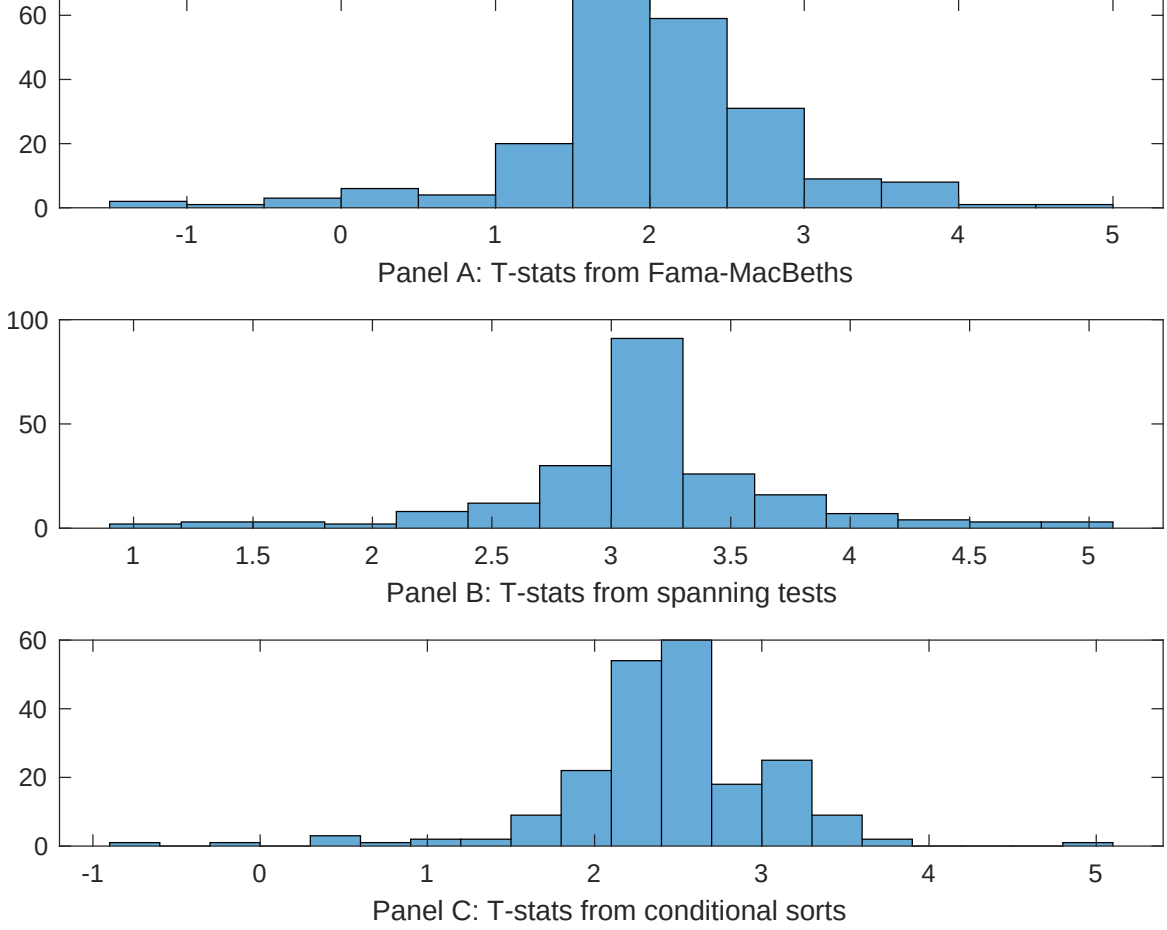




**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ECR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ECR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ECR}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ECR} ECR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ECR,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECR. Stocks are finally grouped into five ECR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECR trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

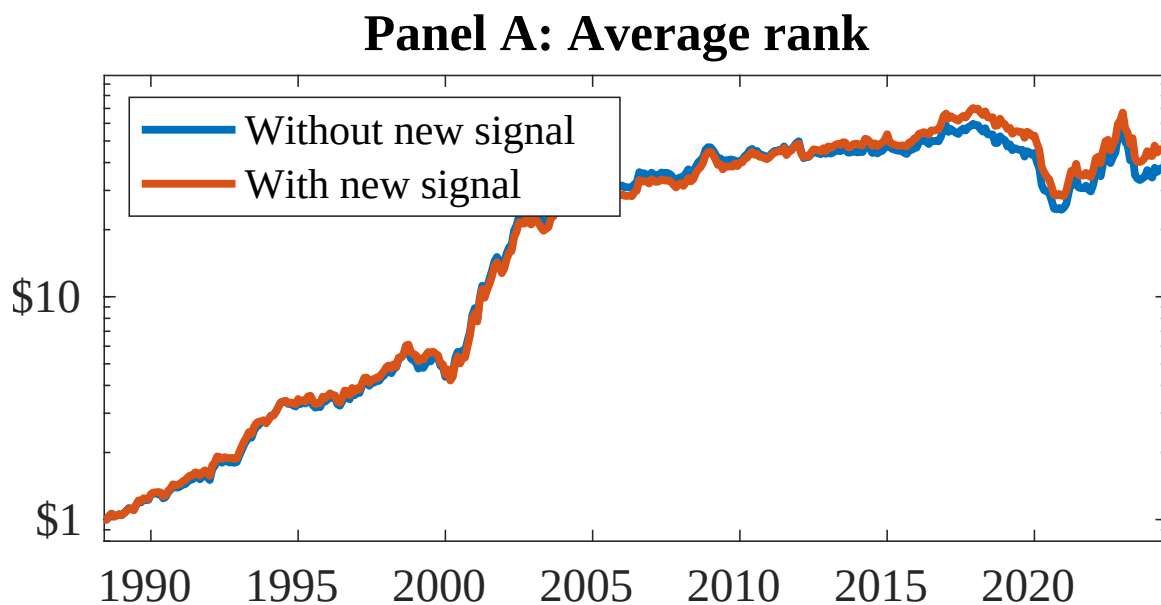
This table presents Fama-MacBeth results of returns on ECR. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{ECR} ECR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

|                 |                |                  |                |                |                |                |                  |
|-----------------|----------------|------------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.90<br>[3.02] | 0.10<br>[3.47]   | 0.87<br>[2.53] | 0.12<br>[4.25] | 0.11<br>[4.73] | 0.11<br>[3.99] | 0.11<br>[5.10]   |
| ECR             | 0.22<br>[2.56] | 0.26<br>[2.49]   | 0.25<br>[3.22] | 0.68<br>[0.07] | 0.25<br>[3.88] | 0.69<br>[0.99] | 0.15<br>[2.16]   |
| Anomaly 1       | 0.22<br>[2.96] |                  |                |                |                |                | 0.23<br>[0.32]   |
| Anomaly 2       |                | -0.84<br>[-0.16] |                |                |                |                | -0.13<br>[-2.35] |
| Anomaly 3       |                |                  | 0.46<br>[0.69] |                |                |                | -0.17<br>[-0.60] |
| Anomaly 4       |                |                  |                | 0.18<br>[6.02] |                |                | 0.11<br>[4.36]   |
| Anomaly 5       |                |                  |                |                | 0.70<br>[1.17] |                | 0.61<br>[0.76]   |
| Anomaly 6       |                |                  |                |                |                | 0.42<br>[2.80] | 0.35<br>[3.09]   |
| # months        | 427            | 432              | 427            | 432            | 427            | 427            | 427              |
| $\bar{R}^2(\%)$ | 0              | 1                | 2              | 1              | 2              | 1              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ECR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{ECR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

|                 |                   |                   |                    |                   |                   |                    |                   |
|-----------------|-------------------|-------------------|--------------------|-------------------|-------------------|--------------------|-------------------|
| Intercept       | 0.21<br>[2.49]    | 0.23<br>[2.57]    | 0.24<br>[2.68]     | 0.20<br>[2.24]    | 0.24<br>[2.68]    | 0.23<br>[2.53]     | 0.19<br>[2.28]    |
| Anomaly 1       | 34.67<br>[8.75]   |                   |                    |                   |                   |                    | 36.13<br>[7.70]   |
| Anomaly 2       |                   | 20.55<br>[3.83]   |                    |                   |                   |                    | -5.03<br>[-0.84]  |
| Anomaly 3       |                   |                   | -2.34<br>[-0.68]   |                   |                   |                    | -17.57<br>[-4.43] |
| Anomaly 4       |                   |                   |                    | 21.40<br>[4.98]   |                   |                    | 11.97<br>[2.57]   |
| Anomaly 5       |                   |                   |                    |                   | 4.07<br>[1.41]    |                    | 5.16<br>[1.61]    |
| Anomaly 6       |                   |                   |                    |                   |                   | 7.05<br>[1.77]     | 5.62<br>[1.40]    |
| mkt             | -3.39<br>[-1.61]  | -4.37<br>[-1.88]  | -7.72<br>[-3.26]   | -4.11<br>[-1.81]  | -5.51<br>[-2.17]  | -6.19<br>[-2.69]   | -3.15<br>[-1.35]  |
| smb             | -29.97<br>[-9.62] | -31.76<br>[-8.82] | -39.78<br>[-10.38] | -31.24<br>[-9.10] | -34.90<br>[-8.64] | -36.95<br>[-11.17] | -32.41<br>[-8.44] |
| hml             | -16.44<br>[-4.50] | -19.61<br>[-5.03] | -20.96<br>[-5.29]  | -19.10<br>[-4.96] | -22.41<br>[-5.61] | -19.00<br>[-4.60]  | -12.21<br>[-3.09] |
| rmw             | 45.21<br>[8.81]   | 58.05<br>[9.44]   | 78.55<br>[14.14]   | 63.13<br>[13.41]  | 71.65<br>[14.12]  | 69.55<br>[12.80]   | 50.00<br>[7.83]   |
| cma             | 23.06<br>[4.33]   | 33.27<br>[5.90]   | 31.59<br>[5.44]    | 17.19<br>[2.77]   | 29.34<br>[5.06]   | 30.46<br>[5.34]    | 17.54<br>[2.93]   |
| umd             | 5.18<br>[2.73]    | 8.27<br>[4.18]    | 9.94<br>[4.68]     | 8.73<br>[4.50]    | 8.50<br>[4.06]    | 7.62<br>[3.42]     | 6.08<br>[2.86]    |
| # months        | 428               | 432               | 428                | 432               | 428               | 428                | 428               |
| $\bar{R}^2(\%)$ | 78                | 74                | 74                 | 75                | 74                | 74                 | 79                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ECR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

# References

- Barro, R. J. and Ursúa, J. F. (2008). Macroeconomic crises since 1870. *Brookings Papers on Economic Activity*, 2008(1):255–335.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Chen, L., Novy-Marx, R., and Zhang, L. (2011). An alternative three-factor model. *Working paper*.
- Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.
- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.
- Cooper, I. and Priestley, R. (2009). Time-varying risk premiums and the output gap. *Review of Financial Studies*, 22(7):2801–2833.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2006). Profitability, investment and average returns. *Journal of Financial Economics*, 82(3):491–518.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Lin, X. and Zhang, L. (2013). The investment manifesto. *Journal of Monetary Economics*, 60(3):351–366.
- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working paper*.
- Papanikolaou, D. and Kogan, L. (2011). Growth opportunities and technology shocks. *American Economic Review*, 101(3):532–536.
- Yang, F. and Kogan, L. (2013). Firm characteristics and stock returns: The role of investment-specific shocks. *Review of Financial Studies*, 26(11):2718–2759.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.